



Software Process Improvement –SEPI-471



Resistance Factors for SPI and SPI ROI

DR. NARGIS FATIMA

ASSISTANT PROFESSOR

DEPARTMENT OF SOFTWARE ENGINEERING , FEC



Agenda

Resistance Factors for SPI

The lecture slides are for reading purpose not allowed to upload on internet/website

Individual-Level Resistance

- **These resistance factors arise from the psychology of the people involved in software development:**

Fear of incompetence: Developers worry new processes will expose skill gaps or make them slower. This is especially common when introducing structured methodologies to experienced but informally-trained practitioners.

Comfort with the status quo: Experienced engineers often feel their current approach works fine. Without evidence of failure, change feels unnecessary and disruptive.

Lack of awareness: Staff may not understand why change is needed. Without communicating the rationale behind SPI initiatives, resistance is a natural response.

Low motivation to change: If there are no incentives or rewards tied to process compliance, individuals will default to familiar habits.

Organizational-Level Resistance

- Organizational resistance stems from the structure, culture, and environment of the company or institution:

Culture of blame: Organizations where failure is punished create environments hostile to experimentation. SPI requires trial and error; a blame culture suppresses this.

Resource constraints: SPI requires time, money, and personnel. When resources are stretched, improvement activities are consistently deprioritized in favor of delivery.

Siloed departments: QA, Development, and Management teams with poor communication lead to fragmented SPI efforts where each unit pursues its own agenda.

Cultural resistance to change: Organizations with deeply rooted traditions or long-established ways of working face the highest inertia when pursuing transformation.

Poor communication: If the goals, benefits, and expectations of SPI are not clearly communicated, skepticism and confusion replace buy-in

Process-Level Resistance

- Some resistance originates from the nature of the proposed processes themselves:

Complexity of proposed changes: If the new process is too complicated or requires extensive documentation, practitioners abandon it during delivery pressure.

Lack of measurement systems: Without metrics, it is impossible to demonstrate improvement, leading to widespread skepticism about whether SPI is working.

Immature baselines: As Deming's principle states, you cannot improve what you cannot measure. Organizations without measurement maturity cannot track progress.

Poorly defined processes: Vague or ambiguous process definitions invite inconsistent adoption, which undermines the entire initiative.

Irrelevance to context: Processes borrowed from large organizations often do not scale down to smaller teams, creating friction and abandonment.

The lecture slides are for reading purpose not allowed to upload on internet/website

Management-Level Resistance

- Management plays a pivotal role in SPI success or failure. Management-level resistance includes
 - Lack of executive sponsorship:** SPI initiatives without active management backing almost always fail. Kotter's change model identifies establishing urgency and visible leadership commitment as foundational first steps.
 - Short-term financial pressure:** The ROI of SPI is long-term; managers operating under quarterly performance pressure resist investments whose returns are measured in years.
 - Unclear SPI goals:** If management cannot articulate what success looks like, teams cannot prioritize the right activities.
 - Absence of a change champion:** A dedicated SPI champion or process improvement group (SEPG) is critical. Without one, initiatives lose momentum and visibility.
 - Conflicting priorities:** When SPI competes with delivery deadlines and no clear prioritization exists, SPI is always the first casualty.



SPI- Return on Investment

SPI Investment: Training, tools, consultants, and dedicated time.

Process Improvement: Adoption of better practices, standards, and measurement systems.

Measurable Outcomes: Fewer defects, reduced rework, faster delivery, better quality.

Financial Returns: Cost savings, revenue from faster market entry, fewer failure-related losses.

Reinvestment: Returns are channeled back into further improvement.

The lecture slides are for reading purpose not allowed to upload on internet/website



Thank you

The lecture slides are for reading purpose not allowed to upload on internet/website